Linear Regression

Wednesday, 30 July 2025 07:02

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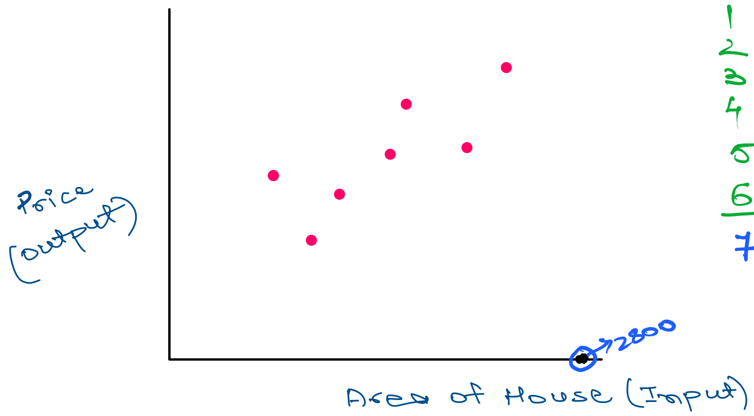
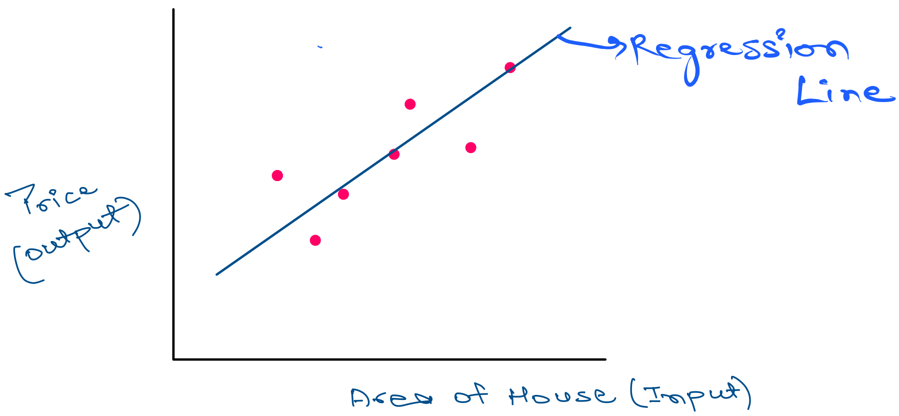
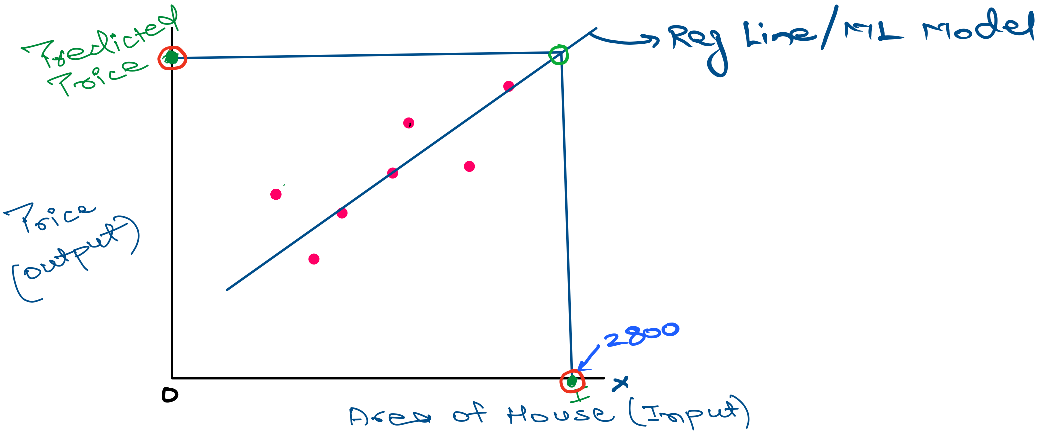
	Area	Price
0	1800	$90k
1	2500	$ 145k
2	4200	$ 280k
3	2100	$ 140k
4	–	–
5	–	–
6	–	–
7	2800	?



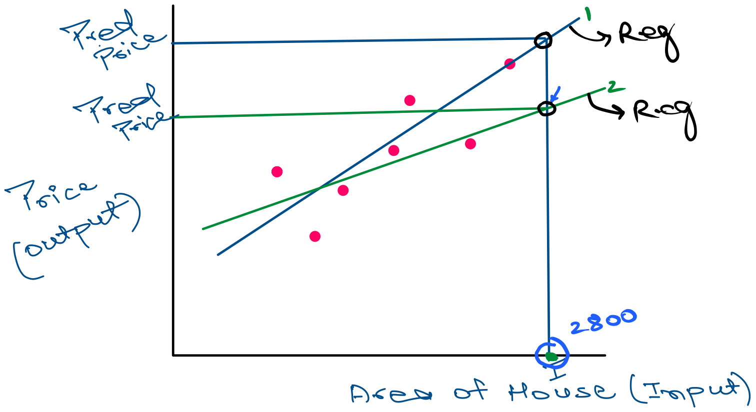
In Linear Regression Algorithm, we need to pass a line through the data such that it is closest to all the datapoints.



Once we have the regression line passing through the data, we can use the line to make predictions.



But there can be many such lines passing through the data and each line would give different prediction. Let's see an example using two such lines.



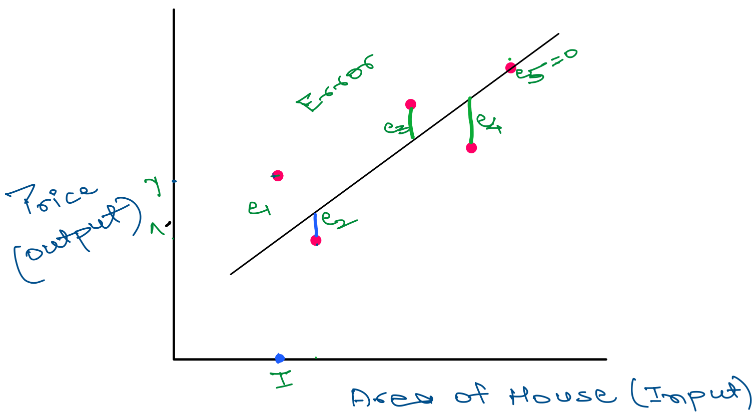
Best Fit Reg Line

As we can observe above, each line gives different predicted output for the same input.

So which line to use then for final prediction?

The regression line we finally use is the one which is closest to the datapoint since that is the line which has learned the relationship b/w input & output with least errors. This line is called 'Best Fit Line'.

Now, what do we mean by an error?



Here,

y = Actual Price

$\hat{y}$ = Predicted Price for the given input (I)

e_i = Error in prediction.

Error = Predicted − Actual

$$e = \hat{y} - y$$

To get the total error made by the line, we add all the errors:

Total errors = $e_1 + e_2 + e_3 + e_4 + e_5$

But here some errors would be positive and some negative and might cancel out each other, so to avoid this issue, we square the errors before adding them.

$$\text{Total Error} = e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2$$

$\downarrow$

We call it <u>sum squared Error (SSE)</u>.

But there is one small problem. If there is a large training data, then SSE will be a very large value and can't be saved in the memory. So we take average of SSE.

$$\frac{1}{n} \times SSE$$

where SSE can be written as,

$$\sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

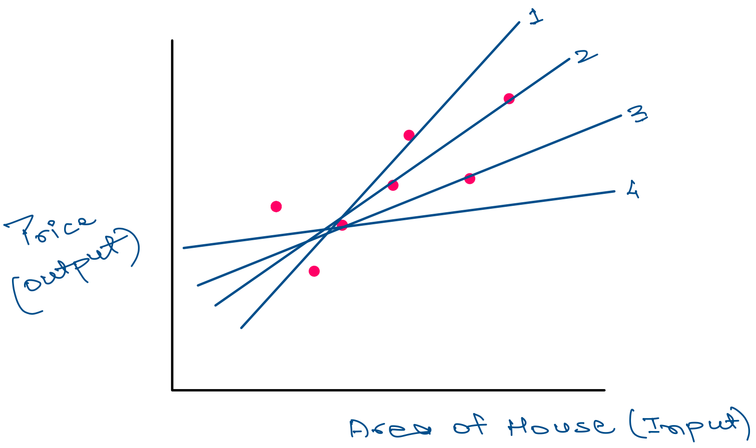
Hence,

$$\frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 \longrightarrow \text{Mean Square Error.}$$

and finally,

$$\boxed{\text{Cost Function } (J(\theta)) = \frac{1}{2n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2}$$

This cost function is used to find the "Best Fit Line". How? Let's see →



To find the Best Fit Line, the algorithm draws ... lines passing through the datapoints and